

KAPIL BHATTARAI

Gainesville, FL | +1 (352) 219-4361 | kapil.bhattarai.161@gmail.com | kbhattarai@ufl.edu | bhattaraikapil.com.np

EDUCATION

M.S. in Agricultural & Biological Engineering

Jan 2025 – Present | GPA: 4.0/4.0

University of Florida, Gainesville, FL — UF Fellowship Recipient | Thesis Advisor: Dr. Gerrit Hoogenboom

B.S. in Agriculture

2017 – 2023

Agriculture and Forestry University (AFU), Rampur, Chitwan, Nepal

TECHNICAL SKILLS

Crop Modeling:	DSSAT v4.8, Model Calibration & Validation, Scenario & Sensitivity Analysis
Programming:	Python, R/RStudio, SQL, Fortran (learning), Git/GitHub, Shiny Apps
Data Science & ML:	Scikit-learn, TensorFlow/TFLite, Predictive Modeling, Time Series Analysis, Data Visualization
Geospatial:	QGIS, GeoPandas, Remote Sensing

EXPERIENCE

Graduate Research Assistant

Jan 2025 – Present

McNair Bostick Simulation Lab, Dept. of Agricultural & Biological Engineering, University of Florida — Advisor: Dr. Gerrit Hoogenboom

- Conducting thesis research on developing a process-based model for lettuce growth under hydroponics settings.
- Developing enhanced graphical visualization tool (GB2) for DSSAT v4.8.6 using Python with modern UI and improved data processing capabilities.
- Building technical skills in Fortran and Python for crop model development and large-scale data analysis.
- Provided technical guidance and DSSAT training to visiting scholars in the Crop Simulation Laboratory.

Nursery Manager

Mar 2024 – Sep 2024

Plant Paradise — Kathmandu, Nepal

- Managed end-to-end nursery operations including plant propagation, production scheduling, inventory control, and retail sales.
- Drove digital marketing (social media, website updates) and applied data-driven sales planning to increase seasonal revenue.

Agriculture Intern

Aug 2022 – Aug 2023

Ragolski & Son's Ltd — Netiv HaAsara, Israel

- Operated precision growing systems in high-tech greenhouses, including coco-peat substrate management, drip irrigation scheduling, and controlled-environment monitoring.
- Gained hands-on experience in hybrid vegetable variety production and commercial cut-flower operations — managing planting schedules, spraying, harvesting, packing, and quality grading.

Undergraduate Researcher & Agriculture Intern

Feb 2022 – Jul 2022

Prime Minister Agriculture Modernization Project (PMAMP) — Chitwan, Nepal

- Precision nitrogen management in spring rice with SPAD, GreenSeeker & leaf color chart; SPAD-guided dosing cut applied N ~33% (120→80 kg/ha) with no yield loss (peer-reviewed, 2024).
- Delivered farmer training on mechanization, vermicomposting, and climate-smart agriculture practices.

SELECTED PROJECTS

DSSAT Graphical Visualization GB2

Python · DSSAT v4.8.6 · Data Visualization · Fortran | 2025 – Present

Enhanced graphical visualization tool for DSSAT crop model outputs, contributing to the global crop modeling research community (100+ countries).

Rogcheck — Crop Disease ID App

CNN · TensorFlow Lite · Java · Android

Android app for offline detection of 10 crop diseases across potato, pepper bell, and tomato using a convolutional neural network.

Personal Portfolio Website

Next.js · React · CSS Modules

Designed and built a full-stack portfolio site with dark/light theme, SEO metadata, and static export hosted at bhattaraikapil.com.np.

CERTIFICATIONS

Certificate in Biological Systems Modeling	University of Florida	9 credit hrs
DSSAT 2025 International Training	University of Georgia, Griffin, GA	May 2025
AI Fluency: Framework & Foundations	Anthropic	2025
Basic Image Classification with TensorFlow	Coursera Project Network	Apr 2022
Remote Sensing Image Acquisition, Analysis & Applications	University of South Wales	Mar 2022
Data Analysis with R Programming	Coursera	Jun 2021
Introduction to Machine Learning	Duke University	Apr 2021
Working with Geospatial Data in Python	DataCamp	Apr 2021
Getting Started with AWS Machine Learning	AWS	May 2021

PRESENTATIONS & EVENTS

DSSAT 2026 International Training — University of Georgia	Faculty	May 2026
Poster Presentation — University of Florida	Hydroponic Lettuce Growth Model	Mar 2026
Crop Physiology Training — Wageningen University & UF	1.5 ECTS Credit	Nov 2025
23rd DSSAT Development Sprint — University of Kentucky	Virtual Participant	Aug 2025
DSSAT 2025 International Training — University of Georgia	72 participants · 28 countries	May 2025
Poster Presentation — University of Florida	DSSAT CSM Performance Evaluation	Mar 2025
HiPerGator Supercomputer Facility Visit — University of Florida	UF Research Computing	Mar 2025
22nd DSSAT Development Sprint — University of Florida	In-person Participant	Jan 2025